Dermatophyte infections, also known as tinea, are the most common fungal **infections** of the skin, hair, and nails. The term "**dermatophyte**" refers to fungal species that infect keratinized tissue, and includes members of the *Trichophyton, Microsporum*, and *Epidermophyton* genera

3. Hair perforation Test

It is used to differentiate *T. mentagrophytes and T. rubrum*. Wedge-shaped perforations in the hair shaft are observed in hair infected with *T. mentagrophytes*.

4. Urease Test

It is used to differentiate between *T. mentagrophytes* and *T. rubrum*. *T. mentagrophytes* hydrolyzes urea and becomes deep red, showing positive result.

iii. Treatment

Whitfield's ointment is used for all Tinea infections. Oral griseofulvin is the drug of choice for nails and scalp infections. Itraconazole and terbinafine may be given as pulse therapy.

9.3 Subcutaneous Mycoses

The fungal infections are characterized by development of lesions at the site of infection by the traumatic inoculation in the subcutaneous tissues. Examples are Mycetoma, Sporotrichosis, Chromoblastomycosis and Rhinosporidiosis.

9.3.1 Mycetoma

Mycetoma is a slowly progressive, chronic granulomatous infection of skin and subcutaneous tissues with involvement of under lying fasciae and bones usually affecting the extremities. Mycetoma is commonly called **Madura foot or Maduramycosis** (Figure 9.7). They are classified into two categories, namely **eumycetoma** cased by **fungi** and **actinomycetoma** caused by higher bacteria of class **actinomycets**.

9.3.2 Pathogenesis and Pathology

The causative agent of Mycetoma is commonly present in saprobic soil source and is transmitted by accidental trauma by thorns or by injury into the subcutaneous tissue. It is common among farmers with minor trauma and abrasions of the skin. Use of wicks for removal of earwax is responsible for Mycetoma of the ear.

HOTS

Is mycetoma occupational disease?



Figure 9.7: Madura foot

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